

Global Outlook

Developing countries projected to use 25% more energy as living standards improve

- >4 billion people currently live in countries where access to energy is below what is needed to address basic human development needs.
- Access to sufficient, affordable energy enables economic development.
- Efficiency gains from new technology help slow energy growth from rising prosperity and a growing population.

Report

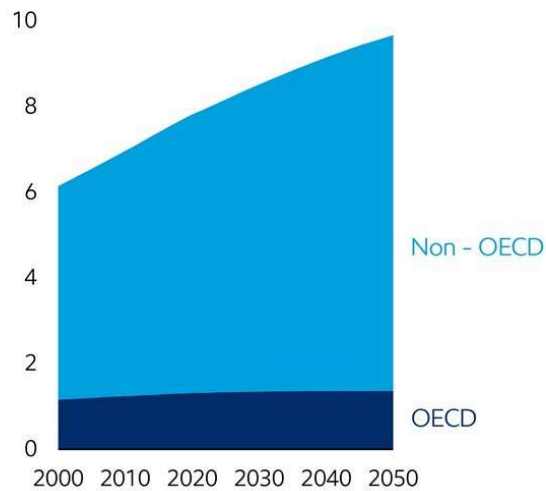
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In 2050, more people and growing economies will drive higher demand for the energy sources that enable modern living, even as efficiency gains enable per capita

energy use to decline in developed economies.

Population

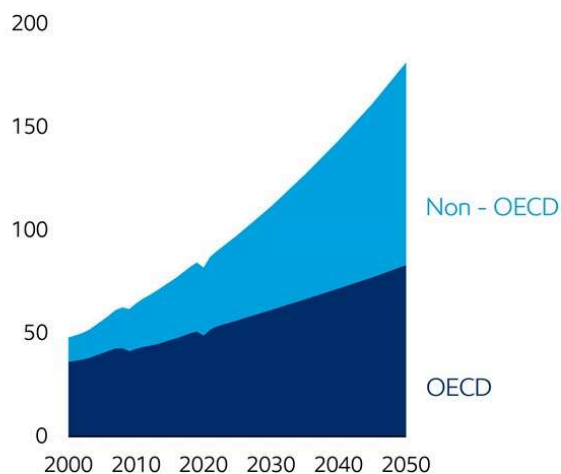
Billions



OECD: Organization for Economic Cooperation and Development, a group of more affluent democracies with market-based economies that promotes economic growth.

GDP

Trillions 2015\$



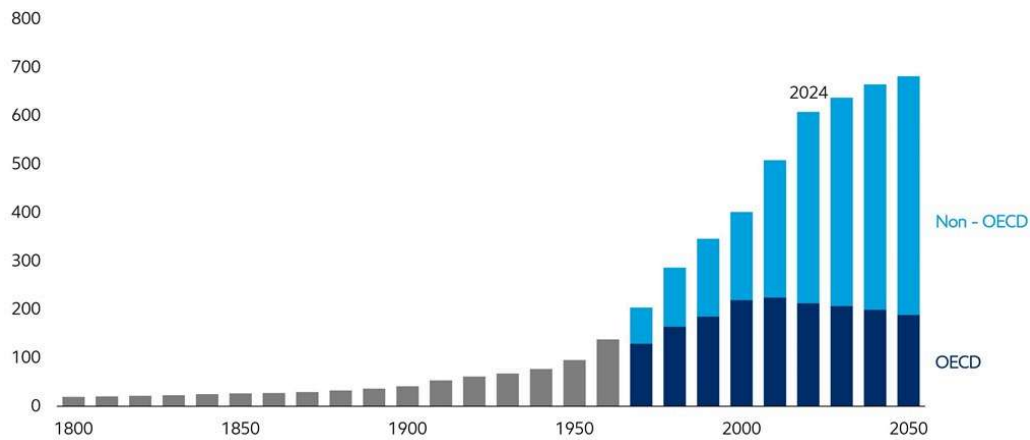
The global population is projected to rise by > 1.5 billion people by 2050, a 20% increase from today, and nearly all of that growth will occur in developing countries.

Over that same time period, global Gross Domestic Product (GDP) is projected to nearly double, with developing nations growing twice as fast as developed nations. By 2050, the developing world will account for more than half of global GDP, up from about 40% today.

The combination of 1.5 billion more people and a global economy that is projected to nearly double in size drives about 25% higher energy use in developing countries in 2050 versus today.

Global energy demand

Quadrillion Btu



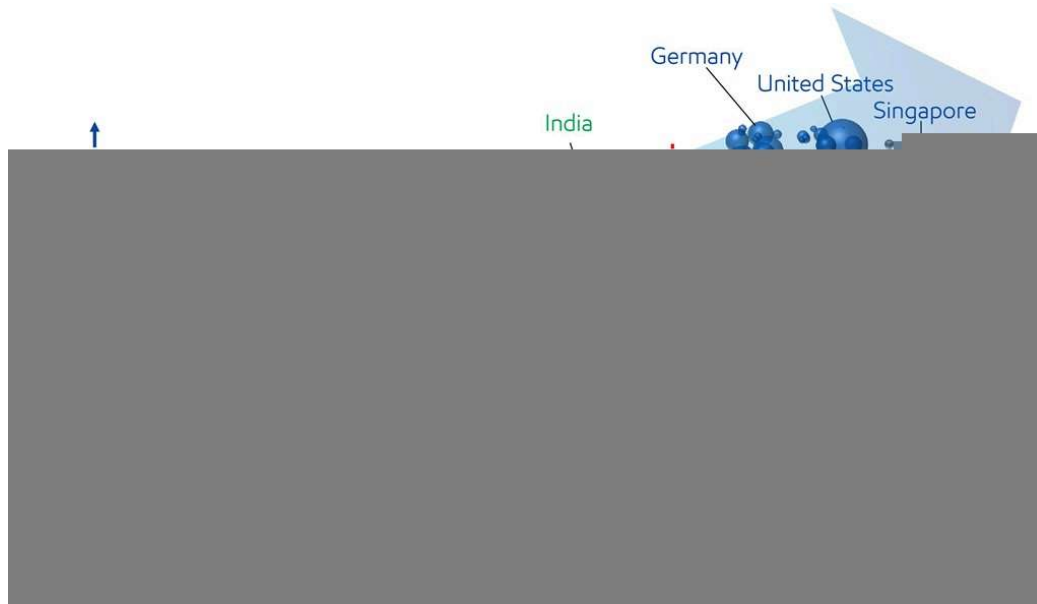
Sources: Smil, Energy Transitions (1900-1960), 2024 ExxonMobil Global Outlook (1970-2050)

Energy use and improved living standards go hand in hand. It is impossible to have one without the other. Data from the United Nations Human Development Index (U.N. HDI) shows that countries with higher energy use tend to have higher life expectancies, education levels, and income per person. Rising energy use fuels

higher incomes that enable people to own homes, purchase labor-saving appliances, travel, and obtain needed medical services.

U.N. Human Development Index

2023 Index



Sources: U.N. Human Development Reports, EIA, ExxonMobil analysis

The U.N. HDI data show us that the opposite is also true: a lack of access to affordable, reliable energy means diminished living standards.

Energy consumption versus access to clean cooking fuels

MMBtu per person per year

Sources: U.N. Human Development Reports, EIA, ExxonMobil analysis

Each symbol denotes a country; bubble size proportional to population

Population weighted averages for UN and World Bank (data through 2024) except energy per capita in MMBtu per person (EIA 2023 data)

Our analysis estimates that basic living standards require at least 50 million British thermal units (MMBtu) per person per year. To put that in perspective, developed countries around the world, on average, use more than three times that amount,

about 160 MMBtu per person, with ~75% of this energy going towards manufacturing, business, and commercial transportation.

Energy use

MMBtu per person per year

Based on our analysis, and the U.N. HDI data, we determined that about 4 billion people live in countries where access to energy is below what is needed to support basic human development, including access to housing, infrastructure, jobs, and mobility.

For example, today in countries with limited access to energy, cooking often involves burning wood, coal, or kerosene indoors. According to the World Health Organization (WHO, Air quality, Energy and Health), more than 3 million people die every year from indoor air pollution, largely associated with burning these fuels. It isn't until a country has access to reliable and affordable energy that households also get access to clean cooking fuels such as electricity or natural gas.

Comparison to third party analyses of minimum energy needs

The Energy for Growth Hub (Moss et al., 2021) has analyzed the level of per capita electricity consumption needed to support modern living standards, and identified the threshold as being 1000 kWh per person per year, with 75% of that being consumed in the wider economy, outside of households. Notably, electricity makes up only about 20% of the world's total energy consumption, with the majority of energy demand being driven by the hard-to-electrify industrial and commercial transportation sectors. Our analysis takes a comprehensive view and considers the total energy mix. Despite this distinction, the analysis of Energy for Growth Hub provides a similar insight as our analysis: today ~4 billion people around the world live in countries where per capita energy and electricity use is less than needed to meet basic human development needs.

The United Nations Sustainable Development Goal 7.1 limits its focus to household "access to electricity", defined by the IEA as at least 50-100 kWh per household per year. It does not consider the broader energy needs outside the home. To put this in context, 50-100 kWh per household per year is sufficient to provide about four hours of electricity per day with a system capable of running basic devices such as lights and phone chargers (IEA, 2025). While access to electricity is a necessary and important first step to addressing energy poverty, this minimum threshold of 50-100 kWh per household per year that is tracked by U.N SDG 7.1 is less than 1% of

the total amount of energy that is needed to address basic human development needs based on our analysis.

Even though developing countries are expected to use significantly more energy through 2050, we predict that total worldwide energy growth will slow as efficiency improves. From 2000 to 2010, total global energy demand grew by 27%. In the past decade, energy demand grew by 13%. Between 2040 and 2050, we project energy demand will grow by less than 3%. Because technology will become more efficient, we expect the next decade will be the first time in history where the world will see expanding economic growth, with lower per capita energy demand.

Energy per capita

MMBtu per person

However, our Outlook also projects limited growth in per capita energy demand in the developing countries, even as overall energy demand in the developing countries grows by 25%.

Sensitivity: How much energy would the world need for all countries to achieve basic living standards (50 MMBtu per person) in 2050? Even with the significant projected efficiency gains, we estimate that providing the energy needed for the

developing world to universally achieve basic living standards would require nearly 20% more energy in 2050 compared to our Global Outlook projection.

Introduction

Our Global Outlook provides important insights on the economic advantages and environmental considerations that play a part in the future of energy.



Industry and commercial transportation drive economic growth

These sectors, which make up ~65% of global energy use, have unique needs that cannot be fully replaced with electricity or renewables.



Learn more about our view to 2050 →